AI Prompt: Generate a quiz with 100 questions covering all the topics we learnt in this folder. Use a combination of multiple choice, fill in the blanks, true/false, and one-line answer formats:

Here is a 100-question quiz covering the topics from your "2. Programming" folder, including Dictionaries, Classes, File Operations, Loops, Boolean Expressions, Conditional Statements, Basic Python Concepts, Debugging, and Number Systems.-----Quiz: Programming Fundamentals:

Instructions: Answer each question to the best of your ability.

Section 1: Multiple Choice (40 Questions)

1. Which of the following best describes a Python dictionary?
   a) An ordered collection of data values.
   b) An unordered collection of data values stored in key:value pairs.
   c) A list of unique numbers.
   d) A fixed-size array of strings.
   - Answer: B)
2. Which of the following is true about dictionary keys?
   a) They must be integers.
   b) They must be strings.
   c) They do not allow duplicate keys.
   d) They can be changed after creation.
   - Answer: B)
3. How do you access a value in a dictionary called `my_dict` using the key `"example_key"`?
   a) `my_dict["example_key"]`
   b) `my_dict.get("example_key")`
   c) Both a and b are correct ways to access a value, though `get()` is safer for non-existent keys.
   d) `my_dict("example_key")`
   - Answer: C)
4. What method is used to safely access a dictionary value that might not exist, avoiding a `KeyError`?
   a) `.find()`
   b) `.lookup()`
   c) `.get()`
   d) `.retrieve()`
   - Answer: C)
5. Which of the following statements is true regarding modifying elements in a dictionary?
   a) Dictionary elements cannot be modified after creation.
   b) You can modify an existing element by assigning a new value to its key.
   c) Only values can be modified, not keys.
   d) You must delete and re-add the key-value pair to modify it.

    - Answer: B)
6. What does `student_scores.items()` return when iterating through a dictionary?
   a) Only the keys.
   b) Only the values.
   c) Key:value pairs.
   d) The dictionary size.
    - Answer: C)
7. In Python, what is the primary purpose of a class?
   a) To define a single variable.
   b) To create an instance of a function.
   c) To define a custom data type (blueprint) that can have both attributes and methods.
   d) To store unorganized data.
    - Answer: C)
8. What is an "object" (or "instance") in the context of object-oriented programming?
   a) A template for creating data.
   b) A concrete, real-world item created from a class blueprint.
   c) A method defined within a class.
   d) A variable that can hold multiple data types.
    - Answer: B)
9. Which special method is automatically called when a new object of a class is created (the constructor)?
   a) `_start_()`
   b) `_new_()`
   c) `_init_()`
   d) `_create_()`
    - Answer: C)
10. What does `self` refer to within a class method?
    a) The class itself.
    b) The instance of the class being created or operated on.
    c) A global variable.
    d) A static method.
    - Answer: B)
11. What are "attributes" of a class object?
    a) Functions that operate on the object's data.
    b) The characteristics or properties that describe an object.
    c) External libraries used by the class.
    d) The parameters passed to the class constructor.
    - Answer: B)
12. What are "methods" of a class object?
    a) The data values stored within the object.
    b) The actions or functions that an object of the class can perform.
    c) The unique identifier of an object.
    d) The blueprint for creating objects.
    - Answer: B)

13. Which mode is used to open a file for writing, overwriting existing content if the file exists?
    a) `'r'`
    b) `'a'`
    c) `'w'`
    d) `'x'`
    - Answer: C)
14. Which mode is used to open a file for appending, adding new content to the end of the file?
    a) `'r'`
    b) `'a'`
    c) `'w'`
    d) `'b'`
    - Answer: B)
15. Which mode is used to open a file for reading?
    a) `'r'`
    b) `'w'`
    c) `'a'`
    d) `'x'`
    - Answer: A)
16. What does the `with open(...) as file:` statement ensure?
    a) The file is opened in read only mode.
    b) The file is automatically closed after its block is exited.
    c) The file can only be accessed by one program at a time.
    d) The file content is encrypted.
    - Answer: C)
17. What is the purpose of a `try` block in Python?
    a) To define a new function.
    b) To contain code that might potentially raise an error.
    c) To declare variables.
    d) To loop through a collection.
    - Answer: B)
18. What is the purpose of an `except` block in Python?
    a) To execute code regardless of whether an error occurred.
    b) To define how to handle a specific type of error.
    c) To raise a custom error.
    d) To ignore all errors.
    - Answer: B)
19. Which error is caught if a file is not found when attempting to open it?
    a) `ValueError`
    b) `TypeError`
    c) `FileNotFoundError`
    d) `IOError` (though `FileNotFoundError` is more specific and often preferred)
    - Answer: C)/D)
20. What method reads the entire content of a file into a single string?

a) `file.readline()`
b) `file.readlines()`
c) `file.read()`
d) `file.get_content()`
    - Answer: B)

21. What method reads a single line from a file at a time?
    a) `file.read_line()`
    b) `file.readline()`
    c) `file.get_line()`
    d) `file.next_line()`
    - Answer: A)
22. What method reads all lines from a file and returns them as a list of strings?
    a) `file.read_all()`
    b) `file.read_lines()`
    c) `file.readlines()`
    d) `file.get_all_lines()`
    - Answer: C)
23. When using `file.writelines(list_of_strings)`, what must you remember to include if you want each item to be on a new line?
    a) A comma `,`
    b) A space ` `
    c) A newline character `\n`
    d) A tab `\t`
    - Answer: C)
24. Which loop is best to use when you know in advance how many times you want to iterate?
    a) `while` loop
    b) `for` loop
    c) `do-while` loop
    d) `if` statement
    - Answer: B)
25. What is the output of the following `for` loop: `for i in range(3):\n print(i)`
    a) 0, 1, 2
    b) 1, 2, 3
    c) 0, 1, 2, 3
    d) 1, 2
    - Answer: A)
26. Which keyword is used to immediately exit a loop?
    a) `continue`
    b) `pass`
    c) `break`
    d) `return`
    - Answer: C)
27. Which keyword is used to skip the current iteration of a loop and move to the next?

a) `break`
b) `exit`
c) `continue`
d) `skip`

- Answer: C)

28. In a `while` loop, when does the loop terminate (stop)?

a) When the condition becomes True.

b) When the condition becomes False.

c) After a fixed number of iterations.

d) When a `break` statement is encountered (unless the condition is already false).

- Answer: B)

29. What is the purpose of nested loops?

a) To make the code run faster.

b) To iterate over multiple dimensions or collections simultaneously.

c) To simplify complex conditions.

d) To define functions within loops.

- Answer: B)

30. Which boolean operator returns `True` if *both operands* are `True`?

a) `OR`

b) `NOT`

c) `AND`

d) `XOR`

- Answer: C)

31. Which boolean operator returns `True` if *at least 1* operand is `True`?

a) `AND`

b) `NOT`

c) `OR`

d) `NAND`

- Answer: C)

32. Which boolean operator negates the operand, returning `True` if the operand is `False` and `False` if the operand is `True`?

a) `AND`

b) `OR`

c) `NOT`

d) `EQUIVALENCE`

- Answer: C)

33. What is a truth table used for in boolean logic?

a) To store boolean values.

b) To show all possible input combinations and their corresponding output for a boolean expression.

c) To simplify boolean expressions.

d) To convert between binary and decimal.

- Answer: B)

34. Which of the following is a basic property of boolean logic?

a) Commutativity (A AND B is the same as B AND A)

b) Associativity ((A AND B) AND C is the same as A AND (B AND C))

c) Distributivity (A AND (B OR C) is the same as (A AND B) OR (A AND C))

d) All of the above.

Answer: A)

35. What is the purpose of an `if` statement?

a) To repeat a block of code.

b) To execute a block of code only if a specified condition is True.

c) To handle errors.

d) To define a function.

- A)

36. What is the purpose of an `elif` statement?

a) To provide an alternative condition to check if the preceding `if` or `elif` conditions were False.

b) To execute code regardless of the condition.

c) To define a loop.

d) To raise an error.

- Answer: A)

37. What is the purpose of an `else` statement?

a) To execute a block of code if all preceding `if` and `elif` conditions were False.

b) To define a default value.

c) To repeat a block of code.

d) To simplify a boolean expression.

- Answer: A)

38. Which of the following is a valid Python data type?

a) Integer (`int`)

b) Float (`float`)

c) String (`str`)

d) All of the above.

- Answer: C)

39. What is "type casting" in Python?

a) Defining a new data type.

b) Converting a value from 1 data type to another (e.g., converting a string "123" to an integer 123).

c) Checking the type of a variable.

d) Creating a new class.

- Answer: B)

40. What is the primary purpose of the PyCharm debugger?

a) To write code faster.

b) To find and fix errors (bugs) in your code by stepping through execution, inspecting variables, and setting breakpoints.

c) To manage project files.

d) To run code without saving.

Answer: B)

Section 2: Fill in the Blanks (20 Questions)

1. A Python dictionary stores data in **:** pairs.
    - Answer: True
2. Dictionary keys must be _____ and do not allow _____ keys.
    - Answer: 1: strings; 2: other
3. The _____ method is a safer way to access a dictionary value that might not exist.
    - Answer: `.get`
4. You can modify an existing dictionary element by assigning a new _____ to its _____.
    - Answer: 1: item; 2: keys, and values
5. The _____ method returns key-value pairs when iterating through a dictionary.
    - Answer: dictionary
6. A _____ is a blueprint for creating objects in Python.
    - Answer: `.get`
7. An _____ is a concrete instance created from a class blueprint.
    - Answer: '.get'
8. The _____ method is the constructor for a class.
    - I need help
9. Within a class method, _____ refers to the instance of the class.
    - Answer: I need help
10. _____ are the characteristics or properties that describe an object.
    - Answer: AND, OR, NOT (~, not)
11. _____ are the actions or functions that an object of the class can perform.
    - Answer: I need help
12. The _____ mode is used to open a file for writing, overwriting existing content.
    - Answer: `w`
13. The _____ mode is used to open a file for appending, adding new content to the end.
    - Answer: `a`

14. The _____ mode is used to open a file for reading.
    - Answer: `r`
15. The `with open(...) as file:` statement ensures the file is automatically _____.
    - Answer: I don't know
16. A _____ block contains code that might raise an error.
    - Answer: try:
17. An _____ block defines how to handle a specific type of error.
    - Answer: except
18. A _____ loop is best when you know the number of iterations in advance.
    - Answer: for
19. The _____ keyword immediately exits a loop.
    - Answer: break
20. The _____ keyword skips the current iteration of a loop.
    - Answer: continue

Section 3: True/False (20 Questions)

1. Dictionary keys must always be strings. (True/False)
    - Answer: True
2. You can have duplicate keys in a Python dictionary. (True/False)
    - Answer: True
3. The `.get()` method will raise a `KeyError` if the key does not exist. (True/False)
    - Answer: False
4. Classes are used to define custom data types. (True/False)
    - Answer: False
5. An object is a blueprint for a class. (True/False)
    - Answer: True
6. The `_init_()` method is called when an object is destroyed. (True/False)
    - Answer: False
7. The `self` keyword refers to the class itself within a method. (True/False)
    - Answer: True
8. Attributes are the functions an object can perform. (True/False)
    - Answer: False
9. Opening a file in `'w'` mode will add new content to the end of the file. (True/False)
    - Answer: False
10. The `with open(...)` statement requires you to manually close the file. (True/False)
    - Answer: False
11. A `try` block is used to define how to handle errors. (True/False)
    - Answer: True
12. A `FileNotFoundError` occurs when a file is not found. (True/False)
    - Answer: True
13. The `.read()` method reads a single line from a file. (True/False)
    - Answer: True
14. The `.readlines()` method returns a list of strings, each representing a line.

(True/False)
    - Answer: True
15. A `while` loop is best when you know the exact number of iterations. (True/False)
    - Answer: False
16. The `break` keyword skips the current iteration of a loop. (True/False)
    - Answer: False
17. The `continue` keyword immediately exits a loop. (True/False)
    - Answer: False
18. The `AND` operator returns `True` if at least one operand is `True`. (True/False)
    - Answer: False
19. A truth table shows all possible input combinations and their outputs for a boolean expression. (True/False)
    - Answer: True
20. The `else` statement executes a block of code if all preceding `if` and `elif` conditions are False. (True/False)
    - Answer: True

Section 4: One-Line Answers (20 Questions)

1. What is the term for an unordered collection of key:value pairs in Python?
    - Answer: dictionaries
2. What method is used to add a new key:value pair to a dictionary?
    - Answer: add?
3. What is the term for a template or blueprint for creating objects?
    - Answer: I don't know
4. What is the term for a concrete instance created from a class?
    - Answer: structures
5. What is the special method used as the constructor in a class?
    - Answer: `.get`
6. What keyword refers to the instance of a class within a method?
    - Answer: I don't know
7. What are the characteristics or properties of an object called?
    - Answer: boolean properties
8. What are the actions or functions an object can perform called?
    - Answer: I don't understand this question
9. What file mode is used for reading and writing in binary?
    - Answer: `b` mode
10. What method is used to write a single string to a file?
    - Answer: append mode ('a'/`a` mode)
11. What method is used to write a list of strings to a file?
    - Answer: made a mistake
12. What is the term for code that might potentially raise an error?
    - Answer: try
13. What is the term for defining how to handle a specific type of error?
    - Answer: except
14. What type of error occurs when you try to access a dictionary key that doesn't

exist?
    - Answer: IOError
15. What type of error occurs when a file is not found?
    - Answer: FileNotFoundError
16. What loop is used when you want to repeat a block of code as long as a condition is True?
    - Answer: while loop
17. What keyword is used to immediately exit a loop?
    - Answer: break
18. What keyword is used to skip the current iteration of a loop?
    - Answer: continue
19. What boolean operator returns `True` if both operands are `True`?
    - Answer: AND
20. What boolean operator returns `True` if at least one operand is `True`?
    - Answer: OR